TRAFFIC FLOW PREDICTION

INTRODUCTION TO MACHINE LEARNING

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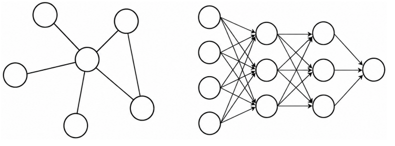
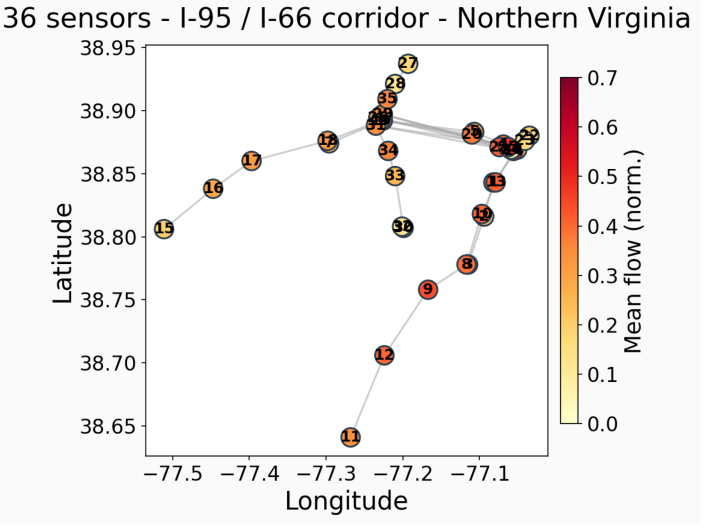


The Dataset

- **Spatial:**
  36 sensors



Source image previous slide: ChatGPT GPT-5.5.

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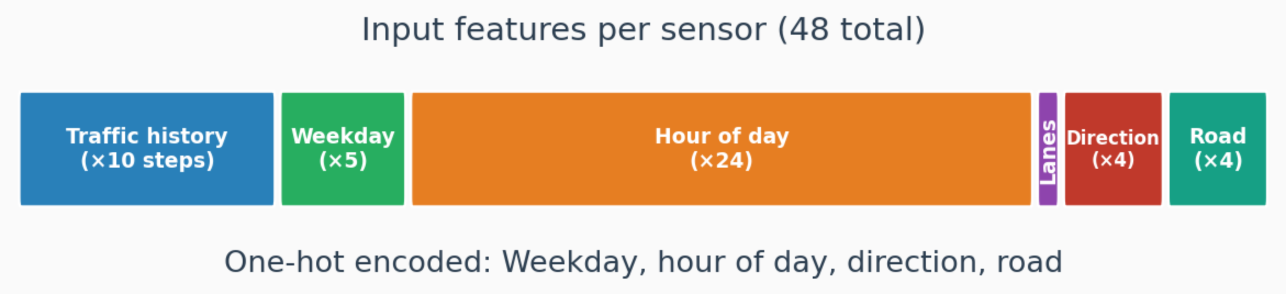
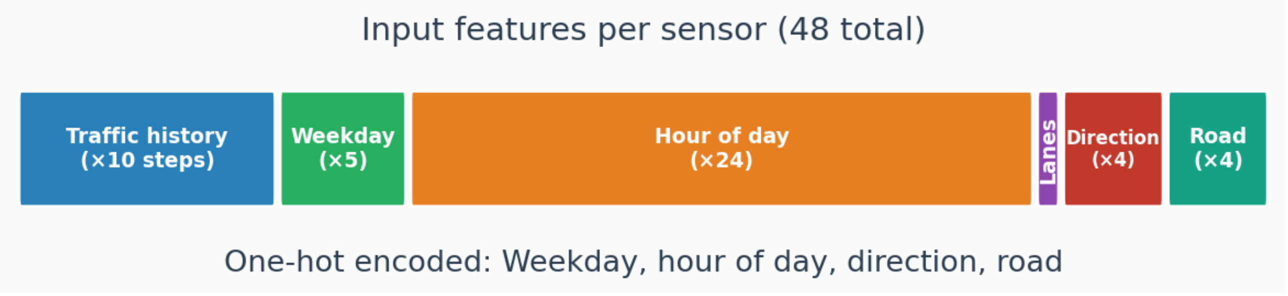


The Dataset

- **Spatial:**
  36 sensors
- **Temporal:**
  2101 time steps x 15 min
- **Feature space:**
  48 features

- **Training set:**
  $1261 \times 36 \times 48$
- **Test set:**
  $840 \times 36 \times 48$
- **Validation set:**
  20% from train (stratified)



The Task

Given **10 past steps** (2.5 h)

$\downarrow$

Predict **next 15 min** on all 36 sensors

Common Training Setup

- Huber loss (robust to outliers, multimodal traffic patterns)
- Feature normalization

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- Feature normalization
- Adam optimizer with weight decay regularization
- Early stopping
- Hyperparameter sweeps across seeds
  (hidden size, dropout, LR)

Model components

Spatial:

- GCN:
  - Similar to CNNs
  - Learns traffic patterns from neighbor sensors
  - “Who are my neighbors, what are they doing?”

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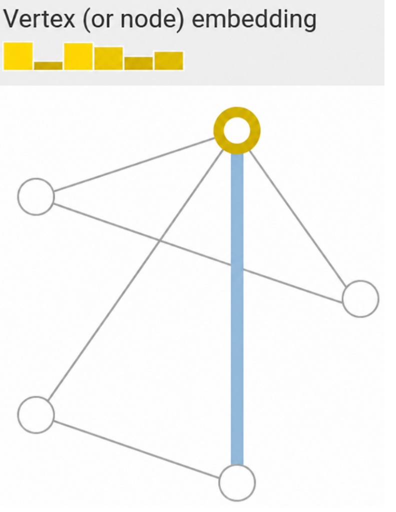
- Sensor embeddings:
  - Learned vector repr. of sensor → [0.2, -0.8, ...]
  - Learns sensor-specific behavior
  - Similar traffic: Highway vs suburban
  - ”Who am I?”



Source: https://distill.pub/2021/gnn-intro/, as of 17.5.2026

Model components

Spatial:

- GCN: Learns traffic patterns from neighboring sensors
- Sensor embeddings: Learns sensor-specific behavior

Temporal:

- GRU: Sequential processing of time steps, memory of past states
- 1D-CNN: Short-term temporal trends/patterns

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Other:

- MLP: Learns non-linear feature interactions
- Transformer: Uses attention to model spatial and temporal dependencies

Models Explored

Baseline

Persistence (copy last)

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Simple

Pure MLP (+ sensor embedding)

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Attention-based

Transformer + cross-sensor

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Transformer + cross-sensor

Graph-based

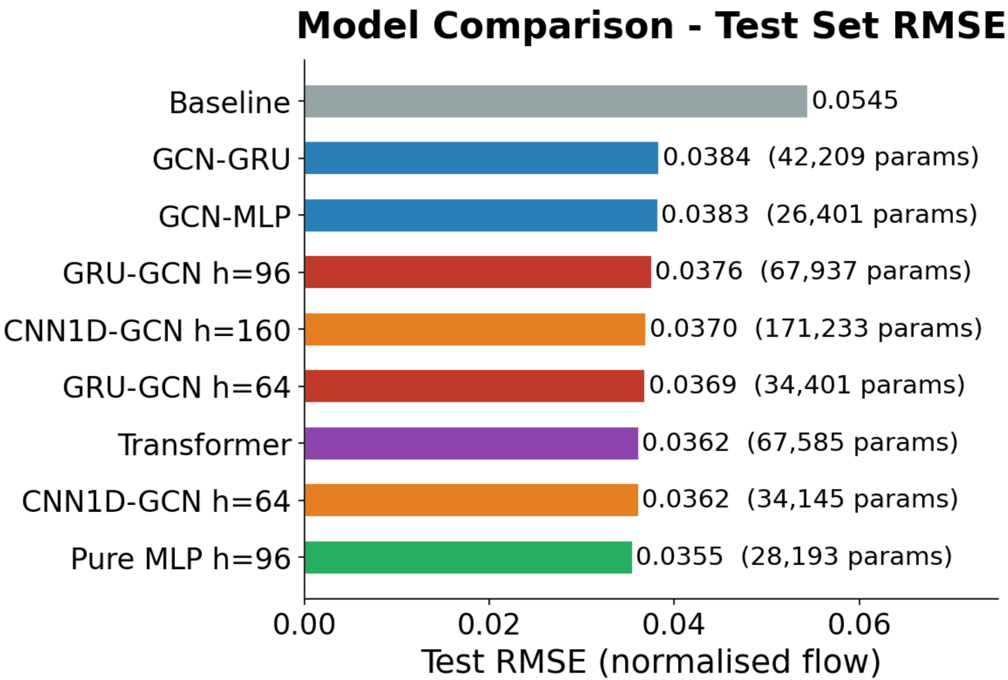
GCN-MLP

GCN-GRU

GRU-GCN (h=64, h=96)

CNN1D-GCN (h=64, h=160)

Results - Test RMSE



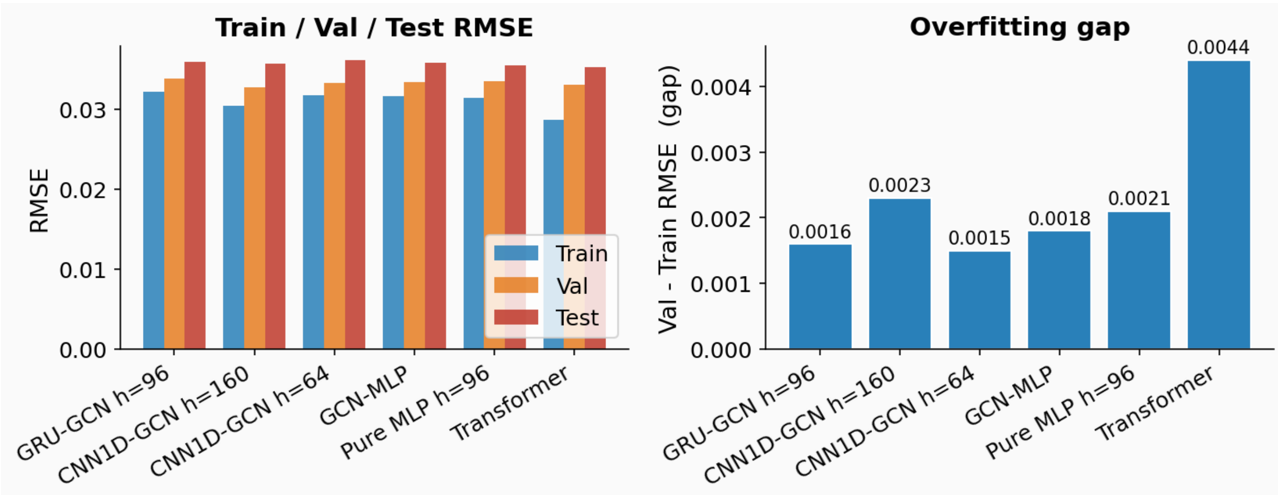
Act 1 - The Climb

Model	Test RMSE	Params
Persistence (baseline)	0.0545	-
GRU-GCN h=64	0.0369	34k
GRU-GCN h=96	0.0376	68k
CNN1D-GCN h=160	**0.0370**	171k

More complex $\rightarrow$ better RMSE

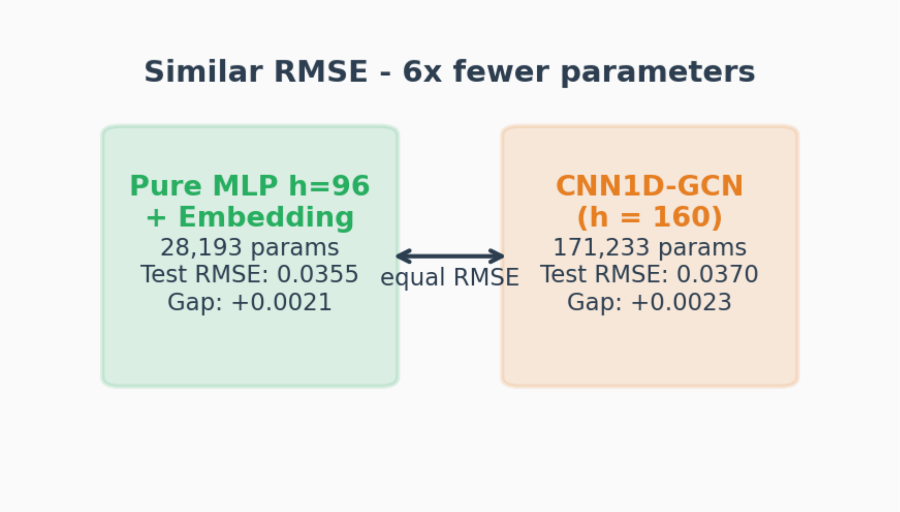
...or is it?

Act 1 - Red Flag



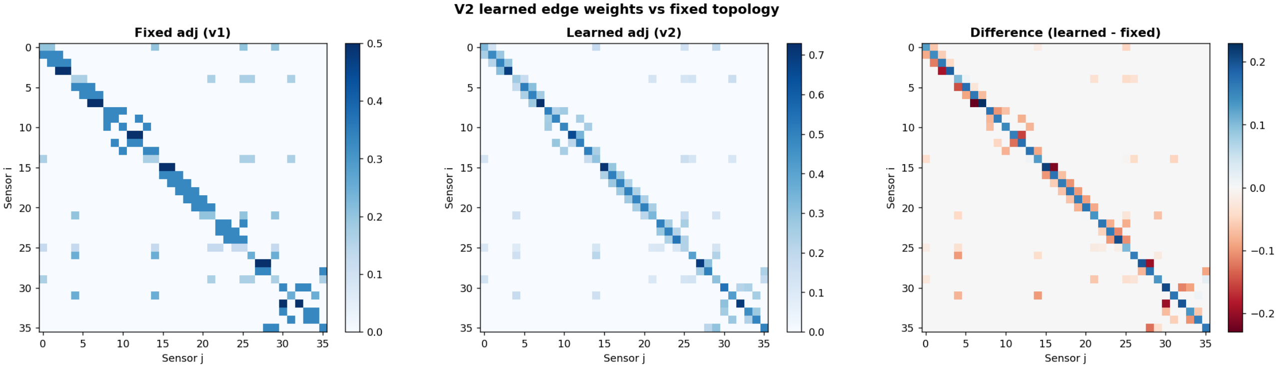
CNN1D-GCN h=160: val gap **+0.0023**, 171k params

Act 2 - The Surprise



Pure MLP h=96 (28k) $\approx$ CNN1D-GCN h=160 (171k)

Why? - Learned Graph is Nearly Uniform



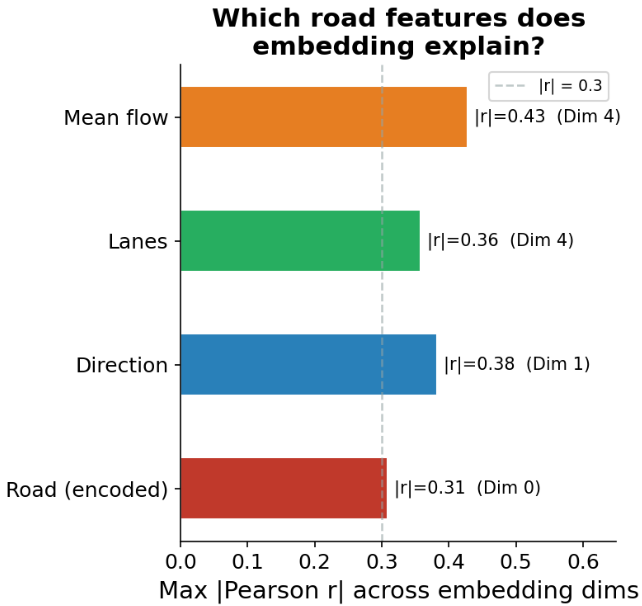
Edge weights 0.35 to 0.65, no strong spatial signal learned

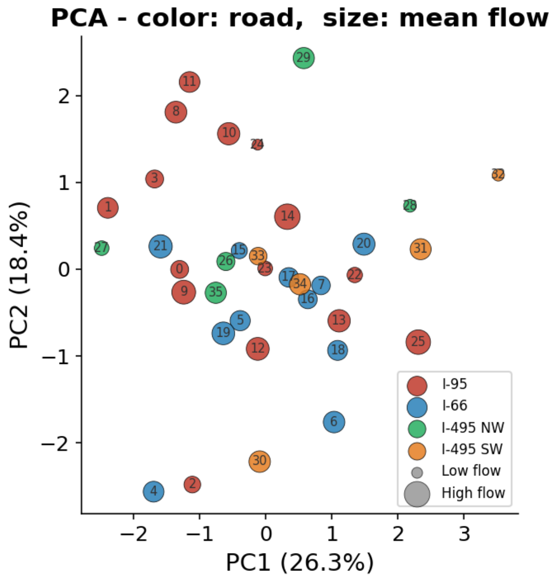
Act 2 - Transformer Confirms It

Model	Test RMSE	Spatial mechanism
Pure MLP h=96	**0.0355**	none (embedding only)
Transformer	0.0362	cross-sensor attention
CNN1D-GCN h=160	0.0370	learned adjacency

Data-driven attention finds no stronger spatial signal than a fixed graph - or than no graph at all

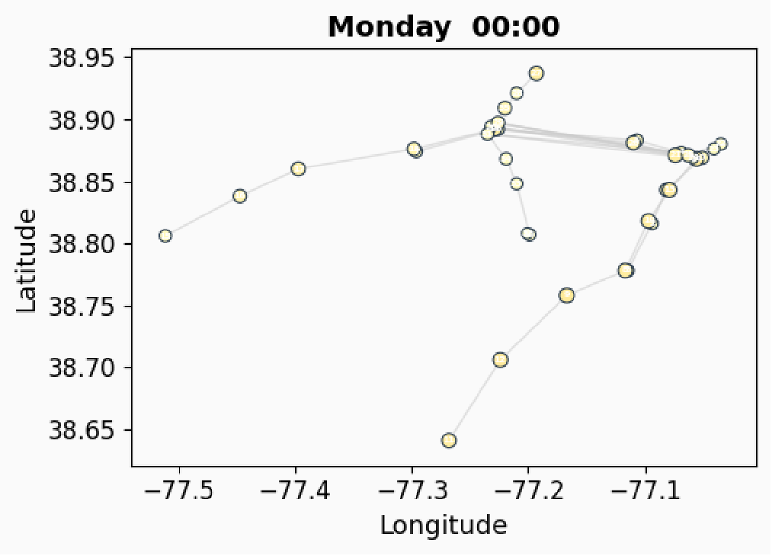
What the Embedding Learned





Dim 4: mean flow + lanes ($r \approx -0.43$) Dim 1: direction ($r \approx -0.38$) 321

A Monday - 24 h of Traffic



Key Points

- **+** Sensor embedding $>$ graph structure
- **+** Time features carry most signal
- **!** Val $\approx$ Train $\neq$ no overfitting
- **!** Capacity / Parameters $\neq$ Performance

The end

Thank you for your attention!

Questions?